

1. Let $N = |T|$ be the tag space. In the case of SHA256, $N = 2^{256}$. Calculating the $\Pr(\text{no collision})$ after trying n times.

$$\Pr(\text{no collision}) = \left(\frac{N-1}{N}\right) \left(\frac{N-2}{N}\right) \cdots \left(\frac{N-n+1}{N}\right)$$

$$= \prod_{i=1}^{n-1} \left(1 - \frac{i}{N}\right)$$

$$\leq \prod_{i=1}^{n-1} e^{-i/N} \quad \text{because } 1-y \leq e^{-y}$$

$$= e^{\sum_{i=1}^{n-1} \frac{-i}{N}} = e^{\frac{-n(n-1)}{2N}} \leq e^{\frac{-(n-1)^2}{2N}}$$

Therefore, the probability of finding a collision satisfies:

$$\Pr(\text{collisions}) = 1 - \Pr(\text{no collisions})$$

$$\geq 1 - e^{\frac{-(n-1)^2}{2N}}$$

$$e^{\frac{-(n-1)^2}{2N}} \leq 1 - x \Leftrightarrow \frac{-(n-1)^2}{2N} \leq \ln(1-x)$$

$$\Leftrightarrow (n-1)^2 \geq 2 \ln\left(\frac{1}{1-x}\right) \cdot N$$

$$\Leftrightarrow n \geq \sqrt{2 \ln \left(\frac{1}{1-x} \right) \cdot \sqrt{N}} + 1$$

Substituting x and N in above formula:

$$x = 0.15, \quad N = 256$$

$$n \geq \sqrt{2 \ln \left(\frac{1}{1-0.15} \right) \cdot \sqrt{2^{256}}} + 1$$

$$n \geq \sqrt{2 \ln \left(\frac{20}{17} \right) \cdot 2^{128}} + 1$$

$$n \geq \sqrt{2} \cdot (0.4031363659) \cdot 2^{128} + 1$$

$$\approx 1.940 \cdot 10^{38} \text{ ans.}$$